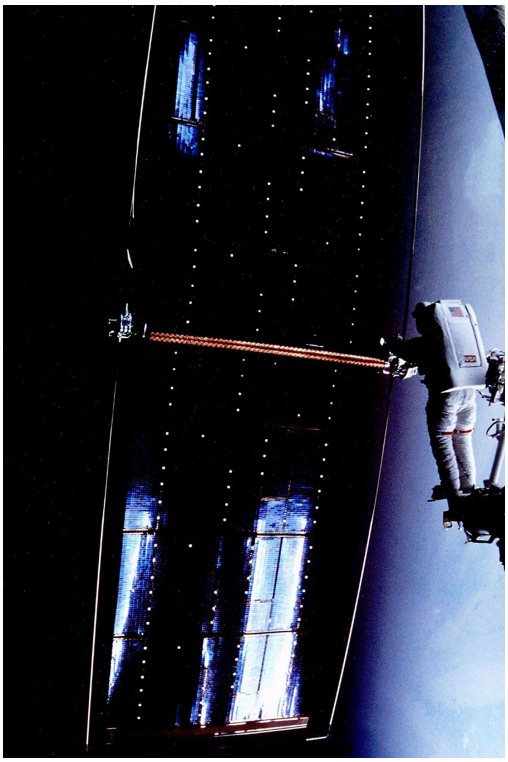
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Hubble's thermal jitter: the missing time scale



Learning objective

Distinguish a slowly varying thermal environment from its dynamic forcing, estimate the residual vibration from a finite-rise load, and identify the evidence needed to protect pointing performance.

A structure can survive launch and still fail its operational vibration requirement. For a precision observatory, small motions can interrupt useful observations long before strength or fatigue becomes the limiting concern.

Hubble makes that distinction tangible. Its solar arrays supplied power, but they also formed flexible mechanical paths into the telescope. The relevant reliability question included how consistently the observatory could hold its line of sight.

Opening photograph: Kathy Thornton releases a removed solar array during Hubble Servicing Mission 1, December 5, 1993. It connects the lesson to the hardware replaced in orbit; the photograph does not diagnose the jitter mechanism. Credit: NASA. Photo and caption [4].

What was observed

A 1993 pointing study reported a design stability of 0.007 arcsec RMS and disturbance content around 0.11 and 0.65 Hz. Solar arrays had become the dominant jitter source; onboard control changes brought performance close to the requirement [2]. These are reported historical quantities, not parameters identified from data in this lesson.

The first servicing mission replaced the arrays in December 1993. NASA describes temperature-driven array motion causing occasional loss of Fine Guidance Sensor lock [3]. The lesson is operational reliability: a recoverable performance disturbance can still consume mission capability.

A thermal cycle is not a forcing waveform

Foster and colleagues distinguished two mechanisms: orbital-transition disturbances were attributed primarily to evolving boom thermal gradients; disturbances later in orbital day were thought to involve frictional stick-slip in drums or spreader bars. A sudden geometric snap-through hypothesis was rejected. Preserve those confidence levels when retelling the case [1].

For a generic appendage, a gradient changes curvature and the equilibrium shape. If the equilibrium changes on a time scale comparable with a structural period, inertia prevents the structure from following it exactly. The difference becomes vibration. The orbital period alone therefore says little about the dynamic content of a terminator transition.

Friction adds a separate route. A constrained interface can store strain energy while it sticks, then release some of that energy as it slips. A smooth temperature history can consequently coexist with abrupt mechanical motion. This is a physical possibility to investigate, not proof that every transient is a friction event.

A minimal model

Let q be one modal displacement, q_{eq} its moving equilibrium, f_n its natural frequency, and ζ its damping ratio. For a constant linear mode, prescribed equivalent forcing gives:

$$\ddot{q} + 2\zeta\omega_n\dot{q} + \omega_n^2 q = \omega_n^2 q_{\mathrm{eq}}(t)$$

$$\omega_n = 2\pi f_n, \qquad q_{\mathrm{eq}}(t) = \frac{F(t)}{k}$$

Here F is generalized force and k is modal stiffness, using a consistent coordinate normalization. For a thermal problem, derive that force from thermal strains and constraints; do not treat free expansion as an externally applied force without checking the boundary conditions.

Suppose the equilibrium ramps from zero to Δq in time τ, then remains fixed. Start from rest and temporarily neglect damping. The residual oscillation amplitude A about the final equilibrium is:

$$\frac{A}{\Delta q} = \left| \frac{\sin(\pi f_n \tau)}{\pi f_n \tau} \right|$$

This follows by integrating the oscillator response to a linear ramp. It isolates the rise-time effect without requiring an assumed spacecraft inertia. The formula applies to this ideal waveform and single mode; a curved thermal history or repeated slip sequence needs its own time-domain forcing.

Equal final load, unequal residual motion

Use f_n = 0.11 Hz and Δq = 1 mm. The frequency is representative of the reported low-frequency disturbance band [2]; the displacement and ramp times below are invented teaching inputs. This is not a reconstruction of Hubble telemetry or an estimate of its line-of-sight jitter.

$$T_n = \frac{1}{0.11\ \text{Hz}} = 9.09\ \text{s}$$

$$\tau = 5\ \text{s}: \quad \frac{A}{\Delta q} = \frac{|\sin(0.55\pi)|}{0.55\pi} = 0.572$$

A five-second ramp leaves 0.572 mm of undamped residual motion. A 0.5-second ramp leaves 0.995 mm; a 20-second ramp leaves 0.085 mm. All three reach the same final equilibrium and static load. Their dynamic responses differ because the loading histories differ.

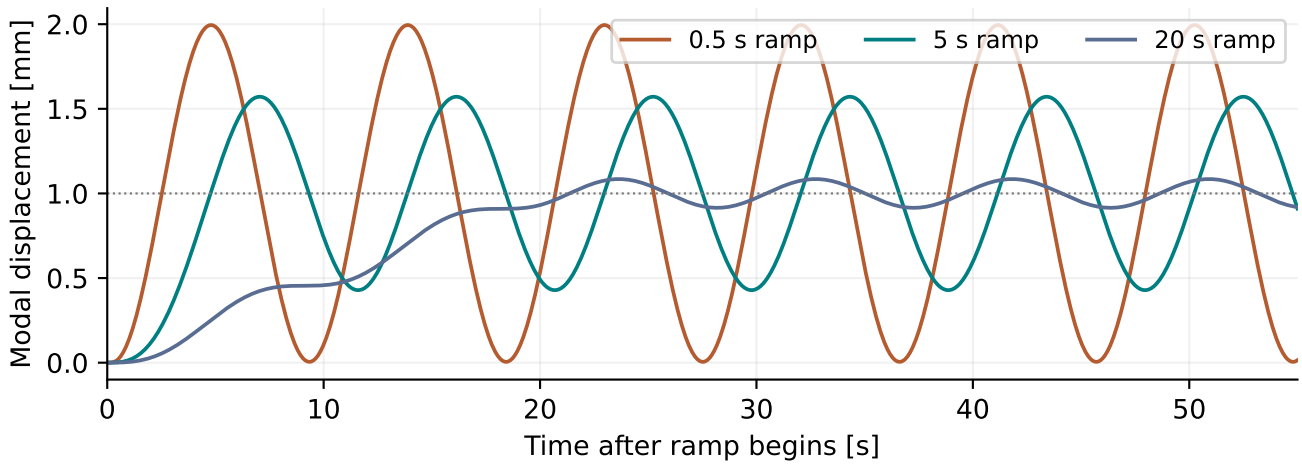


Figure 1. Original analytical calculation for one undamped mode, initially at rest. The dotted line is the common final equilibrium. Curves are illustrative, not measured data. Lack of decay is an explicit modeling assumption.

The slow-ramp benefit is not a robust notch. This ideal expression has exact zeros when the ramp lasts an integer number of modal periods. Small frequency or waveform errors remove that cancellation. Evaluate a range of modal frequencies and plausible transition shapes rather than designing to one favorable zero.

Damping governs persistence. With ζ = 0.01, the free-vibration envelope has an e-fold decay time of approximately 145 seconds at 0.11 Hz. A low-frequency disturbance can therefore outlast the event that excited it. Actual pointing requires the coupled structural and control response, including how each mode projects onto the optical line of sight.

Test the causal chain across disciplines

The engineering approach below is an inference from the physics, not a prescribed Hubble test procedure or a universal standard. Thermal, structural, mechanism and controls teams each own part of the chain. Verification becomes weak when every discipline validates its own model but nobody validates their interfaces.

First, preserve event timing. Align attitude, rates, temperatures, actuator activity and operational state around transitions. Check clocks, latency and sample rate before attributing a phase relationship to physics. A common periodic schedule can create correlation without a direct causal path.

Second, identify response modes separately from the source. A repeated 0.11 Hz response could be the structure ringing after many different inputs. Its frequency alone does not identify a periodic forcing mechanism. Compare onset, decay, spatial response and forcing proxies; seek observations that could falsify the preferred explanation.

Question	Useful evidence
What drives the transient?	Gradient history and constraint model; distinguish temperature from thermal strain.
What releases stored energy?	Mechanism motion, preload and hysteresis; repeatability across thermal cycles.
Which modes carry the response?	Correlated shapes, damping and interface participation; sensor observability.
What reaches the payload?	Closed-loop line-of-sight response, sensor dynamics and actuator limits.

Table 1. Original review aid: evidence linking environment, mechanism, structure and mission performance.

Third, make the experiment representative. A thermal soak verifies an endpoint; it may not reproduce a gradient rate. An ambient modal test identifies a linearized structure; it may not reproduce vacuum friction, thermal preload or flight boundary conditions. Document which part of the causal chain each test actually covers.

Finally, judge the control change against the flexible plant. Disturbance rejection, stability margins, sensor noise and actuator demand must be assessed together. Improved attenuation at one mode can create problems elsewhere. Retain uncertainty in modal frequency and damping instead of treating a correlated nominal model as exact.

Reliability includes useful operating time

The Hubble remedy combined control changes with redesigned arrays. The mechanical changes addressed frictional behavior, while thermal shields reduced boom gradients. The investigation reports satisfactory combined performance after servicing [1]. That is a systems lesson: source reduction and response control can complement each other.

For a new design, begin with the mission metric. A telescope may need a time-windowed pointing error; a guidance package may need bounded rate error; a sensitive instrument may need settling before acquisition. A whole-orbit RMS can dilute a brief but operationally decisive transient. Retain event peaks, settling time and observation-window statistics alongside spectral information.

Strength, fatigue, pointing and control stability are separate acceptance questions. A benign stress response does not establish acceptable pointing, and an acceptable pointing result does not establish structural margin. For safety-critical systems, trace the actual functional consequence through the appropriate hazard analysis; Hubble illustrates performance loss, not evidence of a launch-vehicle breakup mechanism.

Three takeaways

1. Compare time scales. Thermal environments can excite flexible modes when the equilibrium changes fast enough relative to a modal period.

2. Preserve mechanism uncertainty. Distinguish smooth thermal bending, plausible stick-slip and rejected hypotheses; a resonant response does not identify its source.

3. Verify the coupled function. Demonstrate the thermal-to-mechanical-to-control path against the mission metric, with representative transitions and model uncertainty.

Reflection

A redesign halves the final thermal deflection but makes the transition much faster. Is its dynamic response necessarily smaller? **Answer:** no. In the example, halving Δq while changing the ramp from 20 to 0.5 seconds increases residual amplitude from 0.085 to about 0.498 mm. Both amplitude and time history matter.

Sources and scope

[1] C. L. Foster et al., Solar-Array-Induced Disturbance of the Hubble Space Telescope Pointing System, Journal of Spacecraft and Rockets 32(4), pp. 634-644, July-August 1995; NASA-TM-111909. Mechanisms and remedies, especially pp. 641-644. NASA NTRS and full paper.

[2] A. J. Bradley et al., Determination and characterization of the Hubble Space Telescope pointing stability, Spaceflight Dynamics 1993. Historical requirement and frequency bands; NTRS abstract consulted. NASA NTRS.

[3] NASA, Servicing Mission 1 (SM1), mission overview, accessed October 5, 2026. Mission record. [4] NASA, Cast-off: Releasing a Solar Array (1993), release date December 5, 1993. Opening photograph.

Equations, numerical example, plot and review table are original educational analysis. No telemetry is reproduced. No standard is invoked as a universal requirement.